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MILITARY SPECIFICATION

TORCH OUTFIT, CUTTING AND WELDING

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a complete cutting and welding torch outfit suitable for use with acetylene and methylacetylene-propadiene (MAPP) gas when either gas is coupled with oxygen. The outfit may also be addressed as a kit hereinafter.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3433

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SPECIFICATIONS

FEDERAL

H-B-178 - Brush, Wire and Platters, General Specification for.
GG-I-271 - Igniter, Friction, Oxyacetylene-Torch.
ZZ-H-461 - Hose, Rubber, Gas (Acetylene-Hydrogen, Air, and Oxygen).
GGG-H-35 - Hammer, Hand, Welders, Chipping.
GGG-T-558 – Tool Box, Portable, General Specification for.
GGG-T-558/6 - Tool Box, Portable (General, Two Handle Type).

MILITARY

MIL-C-5501 - Gas and Plugs, Protective, Dust and Moisture Seal.
MIL-V-13877 - Valve, Regulating, Acetylene and Oxygen.
MIL-T-13880 - Torches, Oxyacetylene, Cutting and Welding.
MIL-C-17223 - Cleaner, Welding and Cutting Tip.
MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MS15011 - Fittings, Hose, Welding and Cutting, Oxygen and Acetylene.

(Copies of specifications and standards and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other Publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B40.1 - Gages, Pressure Indicating, Dial Type, Elastic Element.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

C4.2 - Operator's Manual for Oxy-Fuel Gas Cutting,

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Component parts. Unless otherwise specified (see 6.2), the portable cutting and welding outfit shall consist of the items and quantity of equipment listed in table I.

TABLE I. Contents of outfit.

Item No.	Item Description	Paragraph reference	Quantity
1	Carrying case.	3.7	1
2	Torch, welding (basic) with reverse flow check valves.	3.8	1
3	Welding tips.	3.9	8
4	Heating tip.	3.9.1	1
5	Other tips.	3.9.2	As specified
6	Cutting tips.	3.9.3	15
7	Gouging tip.	3.9.3	1

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TABLE I. Contents of outfit. - continued

Item No.	Item Description	Paragraph reference	Quantity
8	O-rings.	3.9.10	12 ea type and size.
9	Regulator, acetylene and MAPP.	3.11	1
10	Regulator, oxygen.	3.11	1
11	Gage, acetylene and MAPP.	3.12	1-primary 1-secondary
12	Gage, oxygen.	3.12	1-primary 1-secondary
13	Hose assembly, 25 ft.	3.13	2
14	Hose coupling, male, LH.	3.13.1	1
15	Hose coupling, male, RH	3.13.1	1
16	Coupling nut, remale, LH	3.13.1	2
17	Coupling nut, female, RH	3.13.1	2
18	Coupling tailpieces,straight.	3.13.1	4
19	Ferrules.	3.13.1	6
20	Welding hose crimping pliers.	3.13.1	1
21	Wrench, acetylene valve.	3.14	1
22	Wrench, multipurpose.	3.14	1
23	Cleaners, tip.	3.15	2-sets
24	Igniters, flint, friction.	3.16	4
25	Flint (extra).	3.16.1	24
26	Brush, wire.	3.17	1
27	Hammer, chipping.	3.18	1
28	Operator's manual.	3.21	1

3.2 First article. Unless otherwise specified (see 6.2), the contractor shall furnish, within the time limit specified in the contract, a complete cutting and welding torch outfit for first article inspection. Inspection of

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the first article shall consist of the examination and tests specified in 4.4 and 4.5 which shall be conducted prior to the submission for acceptance of the remainder of the torch outfits furnished under the contract.

3.3 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.3.1 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and the finished product, provided form, fit, and function requirements are satisfied.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term “recovered materials” means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Standard commercial Product. The torch shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer’s standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer’s standard commercial product, shall be included in the torch being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer’s catalogs, or brochures, and represents the latest production model.

3.6 Design and construction.

3.6.1 Design. All parts of the equipment shall be designed for use with oxygen, acetylene, and MAPP gas. All parts shall be of proper size, suitable material, and sufficient strength to ensure satisfactory performance for the cutting and welding services specified.

3.6.2 Threads. Screw threads shall comply with the requirements of FED-STD-H28 unless otherwise specified. All internal and external threads shall be clean, free of any extraneous materials, and shall have no burrs or nicks.

3.6.3 Lubricants. Lubricants on any part of the torch that will be exposed or in contact with oxygen or acetylene and MAPP shall be compatible with oxygen at 90 pound-force per square inch gage.

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3.7 Carrying case. The carrying case shall be designed and constructed to withstand the abuse normal in service without damage to itself or its contents. The case shall conform to GGG-T-558 and type VII size 4 of GGG-T-558/6 exept as modified herein.

3,7.1 Size and weight. The case shall be no larger than 33 inches long, 14 inches wide, 12 inches high, excluding protections.

3.7.2 Handles. The handle opening shall fit the gloved hand. Minimum opening shall be for 5-inch wide and 2-inch thick gloved hand.

3.7.3 Plywood insert. A 1/4-inch thick plywood insert shall be provided.

3.7.4 Tray. A tray shall be provided. The tray shall be equipped with means for easy lifting, and shall be supported in the case in such a manner that its contents will be exposed when the case cover is opened. The tray shall be compartmentized to hold various small components, such as tips, tip cleaners, adapters, etc. , listed in table I. Depth of the tray shall be determined by the manufacturer. The tray shall cover the length and half the width of the inside. Handles or handholds in the ends of the tray shall be provided for ease of lifting the tray out of the case. The tray shall have a bottom, two sides, and two ends.

3.7.5 Coating. Paint the case with olive drab enamel as specified in GGG-T-558.

3.7.6 Quality. Quality assurance for the case shall be in accordance with GGG-T-558.

3.8 Torch cutting and welding.

3.8.1 Design. The torch shall be a handheld positive pressure type, suitable for use with acetylene, MAPP, and oxygen. The torch shall be equipped for controlling and mixing the flow of gases to the torch tip. All parks subject to wear or breakage shall be accessible for adjustment and repair. The torch, cutting attachment, tips, and other parts shall conform to type I of MIL-T-13880 except as modified herein.

3.8.2 Performance. Torches shall be suited for welding 1/2-inch thickness and cutting 6-inch thickness in mild steel base metal without flashback. The torch shall perform as specified above with oxygen and acetylene for welding and cutting, and with oxygen and MAPP for multiflame heating and cutting.

3.8.2.1 Quality of weld. The weld shall be performed with steel rod compatible with the base metal. The filler metal shall flow into an even nonporous molten bead with complete penetration through the base metal. Voids and incomplete fusion of filler and base metal shall not be evident by visual inspection after the slag has been chipped away and the joint wire brushed clean (see 4.5.2).

3.8.2.2 Quality of cut. The cut produced shall be a so called “perfect cut” with a mildly rough surface with slightly sloping drag lines as defined in AWS C4.2. The length of drag shall not exceed 10 percent of metal

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thickness ([Drag percent = drag length divided by metal thickness and multiplied by 100, D percent = [d1 ÷ t] [100]) the kerf width (width of cut) shall not exceed 1.5 times the diameter of the cutting oxygen orifice (1.5d) of the cutting tip (see 4.5,2). The cut shall be a "drop--cut". A drop-cut means the piece being cut off will drop away when the cut is finished (molten slag did not reweld the pieces together).

3.9 Tips. Unless otherwise specified (see 6.2), all tips specified in tables II and III shall be provided. All tips shall conform to all the requirements of MIL-T-13880 except for sizes and amount. All tips shall be manufactured by the same manufacturer at the same plant for the specific torch kit being provided to fill the specific acquisition order.

TABLE II, Welding tips.

Tip orifice size range		
Drill size	Drill diameter	Quantity
70-74	0.0270-0.0210	1
65-69	0.0350-0.0280	1
60-64	0.0400-0.0360	1
55-59	0.0520-0.0410	2
50-54	0.0700-0.0550	1
45-49	0.0820-0.0730	1
40-44	0.0980-0.0860	1

3.9.1 Heating tip. In addition to the welding tips specified in table II a multiflame heating tip and nozzle with a separate mixer, suitable for use with acetylene, and MAPP gas shall be provided. The heating tip shall be of the bent or gooseneck type, having an overall length of not less than 17 inches nor more than 19 inches, shall have not less than 12 orifices of drill size 54-57.

3.9.2 Other tips. When specified (see 6.2), and in addition to the welding tips specified in table II, tips for welding with oxygen and MAPP shall be provided for the drill size number specified in the order. The tips shall otherwise conform to the material and quality requirements of MIL-T-13880.

3.9.3 Cutting and gouging tips. In addition to the cutting tips specified in table 111, a gouging tip suitable for use with acetylene and MAPP gas shall be furnished. The gouging tip shall have a cutting orifice drill size 50-54 and a minimum of six preheat orifices drill size 58-64 and shall be bent at a 30 degree (°) to 40° angle.

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TABLE III. Cutting tips.

Preheat orifice range		Cutting orifice range			
Drill Size	Drill Diameter	Drill Size	Drill Diameter	Gas application with oxygen	Quantity
70-75	0.0280-0.0210	60-65	0.0400-0.0350	Acetylene	1
68-73	0.0310-0.0240	55-59	0.0520-0.0410	Acetylene	1
65-73	0.0350-0.0280	50-54	0.0700-0.0550	Acetylene	1
60-65	0.0400-0.0350	45-49	0.0820-0.0730	Acetylene	1
65-69	0.0350-0.0290	60-64	0.0400-0.0360	MAPP	2
63-68	0.0370-0.0310	55-59	0.0520-0.0410	MAPP	3
60-64	0.0400-0.0360	52-54	0.0635-0.0550	MAPP	2
56-59	0.0465-0.0410	48-51	0.0760-0.0670	MAPP	2
.	slotted	54-56	0.0550-0.0465	Natural gas-propane	1
	slotted	51-53	0.0670-0.0595	Natural gas-propane	1

3.10 O-rings. Six each of replaceable O-rings used as a seal for the gas and oxygen in the welding torch, cutting attachment, tips and mixers shall be provided. The material and size of the O-rings shall be identical to the O-rings installed in the torch and attachments and shall conform to the requirements of MIL-T-13880.

3.11 Pressure regulators. Two regulators shall be supplied, one for use with acetylene and MAPP gas and one for use with oxygen. Each regulator shall be equipped with two pressure gages, one for the cylinder pressure and one for the delivery pressure. All regulators shall be suitable for use with the welding torch described herein while using oxygen and acetylene, or oxygen and MAPP. Unless otherwise specified (see 6.2), the regulators shall be of a two stage design and shall conform to MIL-V-13877, type V for the acetylene regulator, and type VI for the oxygen regulator.

3.12 Gages. The gages shall be suitable for use with oxygen, acetylene, and MAPP as required for the regulators specified in 3.11. Gages shall conform to the requirements of MIL-V-13877, and ANSI B40.1 (accuracy grade A or better) as applicable, except for the features or requirements as modified herein. Each gage shall be built to withstand the rough handling normally encountered in service.

3.13 Hose and couplings. Two 25-foot lengths of 1/4-inch dual oxygen and acetylene hose, "complete with couplings, shall be furnished. The hose shall comply with the requirements of 22-H-461, style B. The coupling tailpiece and coupling nut shall be attached to the hose using crimped ferrules (see figures 1 and 2). The coupling tailpiece shall be the straight stepped type conforming to MS15011.

3.13.1 Hose connection accessories. Hose accessories for making hose to hose couplings, and for use in repairs shall be furnished, and shall consist of the following items in the quantities specified (see figures 1 and 2). Al 1 parts and accessories shall conform to ZZ-H-461, MS15011, and FED-STD-H28 unless otherwise specified.

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a. One hose coupling, male, lefthand threads for acetylene,
b. One hose coupling, male, righthand thread for oxygen.
c. Two hose coupling nuts, female, 1/4 inch, lefthand.
d. Two hose coupling nuts, female, 1/4 inch, righthand.
e. Four coupling tailpieces, straight stepped type, for use with ferrules for 1/4-inch inside diameter hose.
f. Six ferrules for crimping to attach the coupling tailpieces to the 1/4-inch inside diameter hose.
g. One pair of welding hose crimping pliers for 3/16- and 1/4-inch hose of the vise-grip locking type.

3.14 Wrenches. Provide a single wrench to operate the acetylene cylinder valve. Provide a heavy duty multipurpose wrench for:

a. Connecting cutting and welding tips, nozzles, and attachments to the torch.
b. Connecting hoses to regulators and torch body.
c* Connecting regulators to cylinders.

3.15 Tip cleaners. Two sets of tip cleaners to suit all tips listed in tables 11 and III shall be furnished, packed in a metal case of suitable dimensions. The set of cleaners shall conform to the requirements of MIL-C-17223.

3.16 Igniter. Four friction igniters (spark lighters) shall be furnished, each complete with flint. The lighters shall conform to GG-T-271, type I, style A.

3.16.1 Flints. Four boxes of six flints, suitable for use with the spark lighters, shall be provided. The flints, socket, and box shall conform to GG-I-271.

3.17 Brush, wire. Two wire brushes, conforming to H-B-178, type II, class 1, shall be furnished.

3.18 Hammer, chipping. A chipping hammer, conforming to GGG-H-35, type II, class 1 or 2 with style C handle, shall be furnished.

3.19 Treatment and Painting. Treatment and painting shall be in accordance with the manufacturer's best standard practice if not otherwise specified herein or in the referenced specifications.

3.20 Identification marking. The torch outfit shall be marked for identification. The approved item name is "TORCH OUTFIT, CUTTING AND WELDING." The marking shall be placed outside on both sides of the case, in l-inch block letters, in yellow enamel.

3.21 Operator's manual. The contractor shall furnish to the government a standard operator's manual normally furnished in commercial practice for each torch outfit specified in the contract.

3.22 Workmanship. Workmanship shall be of the highest grade throughout and in accordance with the best commercial practice for this type of equipment.

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3.23 Machine work. Tolerances and fitting or matching of connecting or contacting parts shall conform to any limitations specified herein, on the applicable drawings, and otherwise to the standards of best commercial practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First. article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on the torch outfit when a first article is required (see 3.2 and 6.2), This inspection shall include the examination of 4.4, the tests of 4.5, and, when specified, the packaging inspection of 4.6 (see 4.6 and 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

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4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be a complete outfit or kit. All kits offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection, Resubmitted lots shall be reinspected using tightened inspection, If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for examination. Examination shall be based on inspection level II and an Acceptable Quality Level (AQL) of 2.5 percent defective.

4.3.2 Sampling for tests. Tests shall be based on inspection level S-3 and an AQL of 4.0 percent defective.

4.4 Examination. Kits selected for inspection shall be examined and tested to determine conformance with the requirements of section 3 and 5 as indicated in table IV.

TABLE IV. Schedule for inspection.

Examination	Test	Requirement	Paragraph requirement	Paragraph test
X	X	First article.	3.2	4.2.1
X		Interchangeability among items and parts.	3.3	
X		Suitable for MAPP.	3.6.1	
X		Prohibitated lubricants.	3.6.3	
X		Case weight and size.	3.7.1	
X		Case handles.	3.7.2	
X		Plywood insert.	3.7.3	
X		Tray requirements.	3.7.4	
X		Painting of case.	3.7.5	
X	X	Quality of case.	3.7.6	
	X	Torch requirements.	3.8.1	4.5.1
	X	Torch, welding, and cutting.	3.8.2	4.5.2
X	X	Welding, heating, cutting, and other tips in size, type, and amounts as	3.9 thru 3.9.3	4.5.3

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TABLE IV. Schedule for inspection. - continued

Examination	Test	Requirement	Paragraph requirement	Paragraph test
		specified.		
X	X	O-rings as specified.	3.10	4.5.4
X	X	Regulators as specified.	3.11	4.5.4
X	X	Gages as specified.	3.12	4.5.4
X	X	Hose assembly as specified.	3.13	4.5.4
X		Coupling as specified.	3.13.1	4.5.4
X		Coupling nuts as specified.	3.13.1	4.5.4
X		Coupling tailpieces as specified.	3.13.1	4.5.4
X		Ferrules as specified.	3.13.1	4.5.4
X		Crimping pliers as specified.	3.13.1	--
X		Wrench(es) as specified.	3.14	--
X		Tip cleaners as specified.	3.15	4.5.4
X		Igniter as specified.	3.16	4.5.4
X		Flints as specified.	3.16.1	4.5.4
X		Wire brush as specified.	3.17	4.5.4
X		Chipping hammer as specified.	3.18	4.5.4
X		Treatment and painting as specified.	3.19	--
X		Identification as specified.	3.20	--
X		Workmanship as specified.	3.21	--
X		Tolerances, fitting, and matching of connecting, mating, and contacting parts.	3.22	--

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4.5 Tests. Kits selected for test shall be tested as indicated in table IV.

4.5.1 Torch functional test. Conduct the tests of 4.4.1 through 4.4.7 of MIL-T-13880 for the type I torch.

4.5.2 Cut and weld test. The torch shall be connected to an applicable gas dispensing set-up utilizing the gages, regulators, tips, and hose from the kit that the torch is a component of. The torch shall be operated to determine that it can weld two 1/2-inch thick mild steel plates (either butt or bevel joints) together for a sustained length of 6 inches. The torch shall be operated to determine that it can make a smooth cut, in one pass, through a 6 inch thick 12 inch by 12 inch mild steel for a sustained length of 6 inches. The cut and weld shall be performed on a total of three specimens. The weld and cut quality shall conform to the requirements of 3.8.2,1 and 3.8.2.2.

4.5.3 Tips. The welding, heating, and other fuel tips, and cutting tips shall meet the flame test for tips of MIL-T-13880 except that:

a. The tips shall be attached to the torch that belongs to the kit being tested.
b. The tip seat leak test shall be done by plugging the tip end and checking for leaks with soapy water. This is done with the torch and tip at 10 pounds per square inch internal pressure. Tip seats shall not leak when cutting tips are tightened to 20 foot-pounds torque on the tip nut, and welding tips are hand tightened. Plugging the tip shall not apply force between the mating seats.

4.5.4 The following items shall-be tested, or shall have been tested, or shall otherwise meet the inspection and test requirements of their respective specification:

Item name	Available specification
O-rings	MIL-T-13880
Regulators	MIL-V-13877
Gages	MIL-V-13877 ANSI B40.1
Hose assembly	ZZ-H-461 HS15011
Couplings	MS15011
Coupling nuts	MS15011
Coupling tailpieces	MS15011
Ferrules	ZZ-H-461
Tip cleaners	MIL-C-17223
Igniter	GG-I-271
F 1 int	GG-I-271
Wire brush	H-B-178
Chipping hammer	GGG-H-35

4.6 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of section 5.

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5. PACKAGING

5.1. preservation-packaging. The preservation-packaging shall be level A or commercial (C) as specified (see 6.2).

5.1.1 Level A. Each complete torch outfit shall be preserved and packaged in accordance with the level A requirements of MIL-W-45562.

5.1.2 Level C. The complete torch outfit-s shall be preserved and packaged in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The torch outfits shall be packed in accordance with the level A requirements of MIL-W-45562.

5.2.2 Level B. The torch outfits shall be packed in accordance with the level B requirements of MIL-W-45562.

5.2.3 Level C. The torch outfits shall be packed and marked in accordance with ASTM D3951.

5.3 Thread Protection. Threaded components that are not connected onto their mating parts and are subject to thread damage shall be protected from shipping, handling, and storage damage with plastic thread caps or plugs conforming to MIL-C-5501.

6. NOTES

6,1 Intended use. The gas cutting and welding torch outfit is intended for use:

a. With oxygen and acetylene or MAPP gases.
b. In various places requiring the outfit to be portable by two men.
c. In any construction, fabrication, repair, or manufacturing operations, either in-doors or out-of-doors, that requires cutting or welding by hand operation.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Outfit content and quantities, if different (see 3.1).
c. When first article is not required (see 3.2).
d. Tips, other than as specified (see 3.9 and 3.9.2).
e. Regulators, if other than two stage (see 3.11).
f. First article inspection, if different (see 4.2.1).
g. Level of preservation-packaging and level of packing required (see 5.1 and 5.2).

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6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data shall be delivered by the contract or in accordance with the contract or purchase order requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1, The first article should consist of one unit consisting of one complete cutting and welding torch and associated kit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Subject term (key word) listing.

Acetylene torch
Brazing torch
Cutting torch
Hot wrench
MAPP gas
Oxy-acetylene
Torch
Welder
Welding torch

6.6 Changes from Previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:

Navy - YD

(Project 3433-0109)

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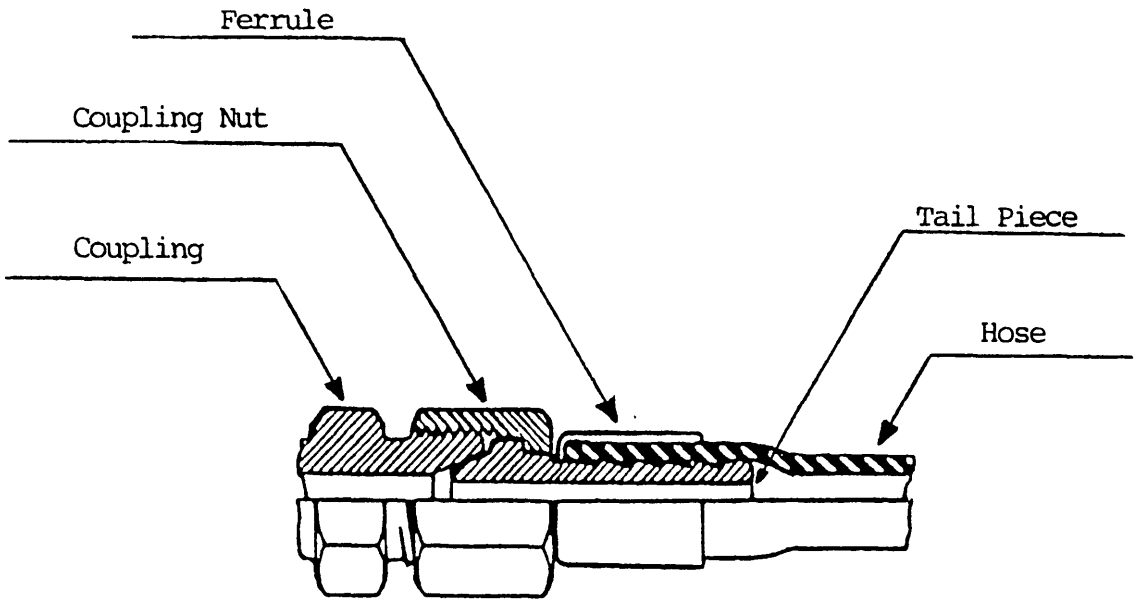


FIGURE 1. Fitted Hose Assembly.

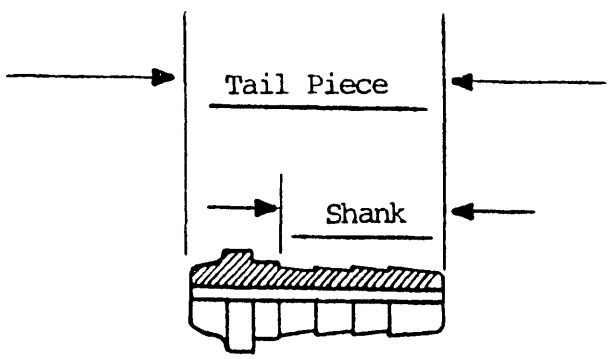


FIGURE 2. Tail Piece With Stepped Shank.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-18803D(YD)	TORCH OUTFIT, CUTTING AND WELDING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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